7.2 Practice: column methods and long division

← Unit 7.2 · Answers

Use squared paper so the digits stay in their columns. Use the method you use at school. Before each sum, you can make a quick estimate (*schatting*): it helps you spot slips.

Set 1 · Warm-up: columns with whole numbers (*cijferend optellen en aftrekken*)

1. 345 + 123 =
2. 576 − 241 =
3. 487 + 235 =
4. 432 − 263 =
5. 2 650 + 1 475 =
6. 5 000 − 1 234 =
7. 12 345 + 8 765 =
8. 30 205 − 4 678 =
9. 600 + 400 =

Talk about it: In question 6, how did you deal with all the zeros? Can you check your answer by adding?

Set 2 · Building: kommagetallen in columns (*komma onder komma*)

1. 2,5 + 1,3 =
2. 4,75 + 2,10 =
3. 6,8 − 2,5 =
4. 12,45 + 3,7 =
5. 10,00 − 3,45 =
6. 25,6 − 8,75 =
7. 3,25 + 0,8 + 12 =
8. At the supermarket you buy bread for € 3,49, cheese for € 2,75 and apples for € 1,20.
   a. How much is that altogether?
   b. You pay with a € 10 note. How much change do you get?
9. Daan cycles 4,6 km to school and then 2,75 km to football. How far does he cycle in total?
10. 1,5 + 1,5 =

Talk about it: In question 7, where does the comma go in “12”?

Set 3 · Building: long multiplication (*cijferend vermenigvuldigen*)

Estimate first. For example, 23 × 46 is about 20 × 50 = 1 000.

1. 3 × 21 =
2. 4 × 536 =
3. 7 × 308 =
4. 12 × 15 =
5. 23 × 46 =
6. 34 × 52 =
7. 125 × 24 =
8. 48 × 37 =

9. The class orders 28 books that cost € 14 each. What is the total?
10. A group of 15 people each buy a train ticket for € 23. What do they pay altogether?
11. 10 × 46 =

Talk about it: Pick one sum. Was your estimate close to the answer? Why was it a bit too big or too small?

Set 4 · Warm-up and Building: dividing with happy numbers (*staartdeling* or *happen*)

All of these come out exactly, with no remainder. Use the way you divide at school.

1. 84 : 4 =
2. 96 : 3 =
3. 848 : 4 =
4. 625 : 5 =
5. 936 : 3 =
6. 672 : 6 =
7. 1 456 : 7 =
8. 2 538 : 9 =
9. 768 : 12 =
10. 1 125 : 25 =
11. 500 : 5 =

Talk about it: How can you check a division? (Hint: multiply.) Try it on question 7.

Set 5 · Stretch: remainders (*rest*) and kommagetallen

1. 25 : 4 = … rest …
2. 47 : 5 = … rest …
3. 826 : 9 = … rest …
4. 126 : 5 = (give the answer as a kommagetal)
5. 50 : 4 = (as a kommagetal)
6. 1 000 : 8 =
7. 130 children go on a school trip. A bus takes 50 children. How many buses are needed?
8. Four friends share a pizza bill of € 30 equally. How much does each friend pay?
9. A farmer packs 100 eggs in boxes of 6. How many boxes can she fill completely? How many eggs are left over?
10. A ribbon of 10 m is cut into 8 equal pieces. How long is each piece?
11. 90 : 3 =

Talk about it: Questions 7, 8 and 9 all have a "leftover". Why is the answer written differently each time?

Puzzle corner (optional)

Digit cards. Use the digits 1, 2, 3 and 4 once each to make a two-digit × two-digit sum: ▢▢ × ▢▢. Which arrangement gives the biggest answer? Which gives the smallest? Can you explain why, without trying them all?

Remainder detective. Find a number between 1 and 50 that leaves remainder 1 when you divide it by 2, by 3 *and* by 4. Can you find more than one? Do you see a pattern?